Bosonization

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1 Introduction

Bosons and fermions are two different beasts. The most important difference between them is the spin. Bosons have integer valued spin, while fermions half integer values. We also know that one cannot put a fermion on top of another, but bosons can cluster together as much as they want to. Despite all these differences, it turns out that in two spacetime dimensions, a theory of free fermions is completely equivalent to a theory of free compact bosons! This is the story of *bosonization*.

This is rather surprising. How the hell is such an equivalence possible? The first thing to notice is that in one spatial dimensions, spin or rotation is a meaningless concept. But spin being meaningless do not solve the difference in statistics of bosons and fermions. How do we disallow bosons to sit on top of one another? We will see in detail that there are solutions in the bosonic theory – that we will call *kinks* or *solitions*– that behaves similar to fermions. We will have kinks and anti-kinks which when superimposed will decay to the vacuum, just like a fermion and anti-fermion.

Bosonization is interesting by itself from the premise. But let us try to give some more motivation as to why this is of interest. Low dimensional quantum field theories like Quantum Electrodynamics in two dimensions (QED_2) are important to describe Condensed Matter systems. Originally, low dimensional field theories were studied for consolation! Gauge theories in $3+1$ dimensions are hard, and often strongly coupled. Our best tool to solve quantum field theory – namely perturbation theory– becomes useless in these cases. We therefore are restricted to numerics in such situations. Properties such as confinement and mass gap of Yang-Mills theories are known only numerically till this day. Low dimensional field theories serve as examples of strongly coupled theories that one can actually solve analytically. We will solve QED_2 in these notes using bosonization procedure, and see that it exhibits many interesting features such as confinement and mass gap. Bosonization is a robust tool that will allow us to solve these strongly coupled theories in two dimensions.

The notes are organized as follows. We start with a quick review of fermions in two dimensions, and then move on to compact bosons in two dimensions. We will discuss T-duality in detail, even though this is a tangent to the main topic. Then bosonization for free theories will be discussed, and lastly will look at two applications, the first one being massive Thirring model dual to sine-Gordon model, and the second is QED_2. References for each section will be mentioned on the way.

2 Fermions in Two Dimensions

In this section, we will study free Dirac fermions in $1+1$ dimensions, given by the action,

$$S = \int d^2x \, \bar{\psi}(i\partial\!\!\!/ - m)\psi \tag{2.1}$$

The conventions we are using are the following. The metric is $\eta_{\mu\nu} = \mathrm{diag}(+1, -1)$, hence the Clifford algebra is,

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \tag{2.2}$$

In particular, we will choose,

$$\gamma^0 = \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \gamma^1 = i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{2.3}$$

We can define the analogue of γ^5,

$$\gamma^5 = i\gamma^0\gamma^1 = \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{2.4}$$

We can use the γ^5 matrix to decompose the two component Dirac fermion field ψ into Weyl spinors,

$$\psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} \tag{2.5}$$

we define,

$$\bar{\psi} = \left(\psi_+^\dagger \psi_-^\dagger\right)\gamma^0 = \left(\psi_-^\dagger \psi_+^\dagger\right) \tag{2.6}$$

The action then becomes,

$$S = \int d^2x \left(i\psi_+^\dagger \partial_- \psi_+ + i\psi_-^\dagger \partial_+ \psi_- - m\left(\psi_+^\dagger \psi_- + \psi_-^\dagger \psi_+\right)\right) \tag{2.7}$$

we have introduced lightcone coordinates,

$$x^\pm = t \pm x, \text{ and } \partial_\pm = \partial_t \pm \partial_x$$

We see that $\psi_\pm$ do not mix if we do not have mass. Let us focus on the massless case. Then, the equations of motion tell us that,

$$\begin{aligned} \partial_+\psi_- = 0 &\Longrightarrow \psi_- = \psi_-\left(x^-\right) \\ \partial_-\psi_+ = 0 &\Longrightarrow \psi_+ = \psi_+\left(x^+\right) \end{aligned} \tag{2.8}$$

ψ_+ (or ψ_-) will be called left (or right) movers. This is because they are function of only $t+x$ (or $t-x$) which just means waves moving to the left (or right) with velocity -1 (or +1).

2.1 Symmetries and Currents

Massive fermions have one global symmetry. It is the usual $U(1)$ symmetry that we are familiar with,

$$\begin{aligned} \psi &\longrightarrow e^{i\alpha}\psi \\ \bar{\psi} &\longrightarrow e^{-i\alpha}\bar{\psi} \end{aligned} \tag{2.9}$$

The conserved current is,

$$j_V^\mu = \bar{\psi}\gamma^\mu\psi \tag{2.10}$$

where the subscript "V" stands for vector. The massless fermions have one additional symmetry,

$$\begin{aligned} \psi &\longrightarrow e^{i\gamma^5\alpha}\psi \\ \bar{\psi} &\longrightarrow \bar{\psi}e^{i\gamma^5\alpha} \end{aligned} \tag{2.11}$$

It is easy to check that this is a symmetry. What it does is to rotate the two Weyl fermions by opposite phases,

$$\begin{pmatrix}\psi_+ \\ \psi_-\end{pmatrix} \to \left(\begin{array}{cc} e^{i\alpha} & 0 \\ 0 & e^{-i\alpha} \end{array}\right)\begin{pmatrix}\psi_+ \\ \psi_-\end{pmatrix} \tag{2.12}$$

Having a mass term will prohibit this symmetry. This symmetry will be called axial, while the other symmetry vector. The conserved current for the axial symmetry is also simple to work out,

$$j_A{}^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi \tag{2.13}$$

As an aside, the following is also indeed a symmetry,

$$\begin{pmatrix}\psi_+ \\ \psi_-\end{pmatrix} \to \left(\begin{array}{cc} e^{i\alpha_1} & 0 \\ 0 & e^{i\alpha_2} \end{array}\right)\begin{pmatrix}\psi_+ \\ \psi_-\end{pmatrix} \tag{2.14}$$

This is a two parameter symmetry, and therefore can be decomposed into a composition of a vector and axial symmetry, each providing one parameter.

2.2 Quantization

Let us take the right mover, ψ_-. Its momentum p is positive. Therefore, the dispersion relation will be,

$$E_p^- = p > 0 \tag{2.15}$$

We can decompose ψ_- into modes,

$$\psi_-(x) = \int_0^\infty \frac{dp}{2\pi}\left(b_-(p)e^{ip\cdot x} + c_-(p)^\dagger e^{-ip\cdot x}\right)_{p^0=E_p^-} = \int_0^\infty \frac{dp}{2\pi}\left(b_-(p)e^{ipx^-} + c_-(p)^\dagger e^{-ipx^-}\right) \tag{2.16}$$

In the second equal sign, we have made use of the dispersion relation to write $p \cdot x = E_p t - px = p(t-x) = px^-$. We see that ψ_- is only a function of x^- as promised. Similarly, for the left mover, the dispersion relation is,

$$E_p^+ = |p| = -p > 0 \tag{2.17}$$

and we have the mode decomposition,

$$\psi_+(x) = \int_{-\infty}^{0} \frac{dp}{2\pi} \left(b_+(p) e^{ipx^+} + c_+(p)^\dagger e^{-ipx^+} \right) \tag{2.18}$$

To quantize, we find the conjugate momenta,

$$\pi_\pm = \mathcal{L} \frac{\overleftarrow{\partial}}{\partial \dot{\psi}_\pm} = i\psi_\pm^\dagger \tag{2.19}$$

and the quantization condition gives,

$$\left\{ \psi_-(x), \psi_-^\dagger(y) \right\} = \delta(x-y) \tag{2.20}$$

We can then derive the algebra for the creation and annihilation operators,

$$\left\{ b_-(p), b_-(q)^\dagger \right\} = 2\pi\delta(p-q), \quad \text{and} \quad \left\{ c_-(p), c_-(q)^\dagger \right\} = 2\pi\delta(p-q) \tag{2.21}$$

Similar relations for the left mover. The left and right movers live on different branches of the dispersion relation, and they anti-commute with each other. The vacuum will be defined to be annihilated by $b_\pm(p)$ and $c_\pm(p)$.

2.3 Correlation Functions

To evaluate the correlation functions, we will introduce a UV regulator in the following form,

$$\begin{aligned} \psi_-(x) &= \int_0^\infty \frac{dp}{2\pi} \left(b_-(p) e^{ipx^-} + c_-(p)^\dagger e^{-ipx^-} \right) e^{\frac{-p}{2\Lambda}} \\ \psi_+(x) &= \int_{-\infty}^0 \frac{dp}{2\pi} \left(b_+(p) e^{ipx^+} + c_+(p)^\dagger e^{-ipx^+} \right) e^{\frac{+p}{2\Lambda}} \end{aligned} \tag{2.22}$$

We can now evaluate two point functions,

$$\begin{aligned} \left\langle \psi_-(x)\psi_-^\dagger(y) \right\rangle &= \int_0^\infty \frac{dp}{2\pi} \int_0^\infty \frac{dq}{2\pi} \left\langle b_-(q) b_-(p)^\dagger \right\rangle e^{-ip\cdot y + iq\cdot x} e^{-\frac{|p|+|q|}{2\Lambda}} \\ &= \int_0^\infty \frac{dp}{2\pi} \int_0^\infty \frac{dq}{2\pi} (2\pi)\delta(q-p) e^{-ip\cdot(x-y)} e^{-\frac{|p|}{\Lambda}} \\ &= \int_0^\infty \frac{dp}{2\pi} e^{-ip\cdot(x-y)} e^{-\frac{|p|}{\Lambda}} \\ &= \frac{i}{2\pi} \frac{1}{(x-y)+i\epsilon}, \quad \text{where} \quad \epsilon = \frac{1}{\Lambda} \end{aligned} \tag{2.23}$$

Similarly,

$$\left\langle \psi_+(x)\psi_+^\dagger(y) \right\rangle = -\frac{i}{2\pi}\frac{1}{(x-y)-i\epsilon} \tag{2.24}$$

As a check, we can evaluate the commutation relations,

$$\begin{aligned}\left\langle \left\{ \psi_-(x), \psi_-^\dagger(y) \right\} \right\rangle &= \frac{i}{2\pi}\left(\frac{1}{(x-y)+i\epsilon} + \frac{1}{(y-x)+i\epsilon}\right) \\ &= \frac{1}{\pi}\frac{\epsilon}{(x-y)^2+\epsilon^2} \to \delta(x-y), \quad \text{when } \epsilon \text{ is small.}\end{aligned} \tag{2.25}$$

and,

$$\left\langle \left\{ \psi_+(x), \psi_+^\dagger(y) \right\} \right\rangle = \frac{1}{\pi}\frac{\epsilon}{(x-y)^2+\epsilon^2} \to \delta(x-y), \quad \text{when } \epsilon \text{ is small.} \tag{2.26}$$

3 Compact Bosons in Two Dimensions

We will consider a single free scalar field in two dimensions, given by the action,

$$S = \int_M d^2x \, \frac{\beta^2}{2} \partial_\mu \phi \partial^\mu \phi \tag{3.1}$$

with one little twist. We will take the field ϕ to take values only in the range,

$$\phi \in [0, 2\pi) \tag{3.2}$$

In other words, we identify $\phi \sim \phi + 2\pi$. For this reason, we call ϕ a *compact boson*[1].

One might wonder why β is kept in front of the action. Usually, we can redefine the field to absorb β, and therefore this constant does not have any physical significance. However, if the boson is compact, then absorbing the constant to the field essentially changes the periodicity of the boson. Therefore, β indeed has a physical significance and we will call it the *radius* of the boson.

To motivate why someone should care about a compact boson, let us turn to string theory. In string theory, ϕ could be interpreted as a direction in the target spacetime. To have bosonic string theory to be anomaly-free, the target spacetime in string theory must be 26 dimensional! However, we only see 4 dimensions in the world. To rectify this issue, people came up with the idea of *compactification*. The idea is to take the extra dimensions, and make them into small circles, so small that we need enormous energies to probe these dimensions. The way one would do this in practice is to take a direction in space that we want to compactify, say X^{25}, and identify $X^{25} \sim X^{25} + 2\pi\beta$, where β is the radius of the circle. The Polyakov action for the string is just (3.1), but with β set to 1. Thus, our compact boson ϕ is then a rescaled version of X^{25}, such that the period is 2π.

Let us also notice that terms like ϕ^n in the Lagrangian will not respect the periodicity of the boson, and hence are not well defined operators. The operators, $\partial_\mu \phi$ or $e^{i\phi}$ are well defined, and we will make use of these soon.

[1]In the strict mathematical sense of the word "compact", the set $[0, 2\pi)$ is non-compact. This is because we can find a cover for this set that does not have a finite sub-cover. We are, as usual, abusing the word a little bit!

3.1 Symmetries and Currents

The action has a constant shift symmetry, $\phi \to \phi + \text{constant}$. This results in the conserved current,

$$j^{\mu}_{\text{shift}} = \beta^2 \partial^{\mu} \phi \tag{3.3}$$

The β^2 is included as a convention. This Noether current is conserved on-shell, which means that the conservation holds whenever ϕ satisfy the equation of motion, $\partial^2 \phi = 0$.

In two dimensions, we get another conserved current for free! This is given by,

$$j^{\mu}_{\text{wind}} = \frac{1}{2\pi} \epsilon^{\mu\nu} \partial_{\nu} \phi \tag{3.4}$$

Notice that this current is conserved due to the anti-symmetry of the ϵ symbol. The field need not be on-shell for this current to be conserved! To explain the subscript "wind", let us take the spacetime manifold also to be compact in the spatial direction,

$$M = \mathbb{R} \times S^1(R), \text{ where } R \text{ is the radius of the circle.} \tag{3.5}$$

Consider an on-shell, static configuration,

$$\partial_x^2 \phi = 0 \Longrightarrow \phi(x) = ax + b \tag{3.6}$$

The wind charge then is,

$$Q_{\text{wind}} = \int_0^{2\pi R} dx \, \frac{1}{2\pi} \epsilon^{01} \partial_1 \phi = \frac{1}{2\pi} \int_0^{2\pi R} dx \, \partial_x \phi = \frac{\phi(2\pi R) - \phi(0)}{2\pi} \tag{3.7}$$

Naively, it seems like the wind charge is always zero, as we would normally identify ϕ at the ends of the circle to have a single valued field. However, due to the periodicity of ϕ, we can relax the single valuedness, and only demand,

$$\phi(2\pi R) = \phi(0) + 2\pi n, \quad n \in \mathbb{Z} \tag{3.8}$$

Then $Q_{\text{wind}} = n$, also, $a = \frac{n}{R}$ so that,

$$\phi(x) = \frac{nx}{R} + b \tag{3.9}$$

We see that n counts how many times ϕ winds around the unit circle as x goes once around the spatial circle of radius R. This is the reason we called the charge and current "wind" charge and current.

3.2 T - Duality

It is useful to bring the string compactification analogy back. If ϕ is a compact dimension of the world where the string can move, then T-duality tells us that the string cannot distinguish if the compact dimension is of radius (proportional to) β or $1/\beta$. We will first derive this duality using path integral techniques, before trying to interpret what this means[2].

Let us start with the partition function (as Lorentzian path integral) of the compact boson theory,

$$Z(\beta) = \int \mathcal{D}\phi \ \exp\left(i \int_M d^2x \frac{\beta^2}{2} \partial_\mu \phi \partial^\mu \phi \right) \tag{3.10}$$

Let us recall that the compact boson can have winding, due to its periodicity. Therefore, the path integral is actually summing over all possible windings as well.

The next step is to gauge the shift symmetry of the action. We impose,

$$\phi(x) \to \phi(x) + \lambda(x) \tag{3.11}$$

as a symmetry. As usual, to make this a symmetry (or rather, a redundancy) of the theory, we should introduce a gauge field, and change the derivatives to covariant derivatives,

$$\partial_\mu \phi \to \mathcal{D}_\mu \phi = \partial_\mu \phi + A_\mu \tag{3.12}$$

Also, the gauge field itself transforms as,

$$A_\mu \to A_\mu - \partial_\mu \lambda \tag{3.13}$$

However, this new gauge theory is not the same as the original compact boson theory. This is a compact boson in an "electromagnetic" field. We will now undo this gauging step, by introducing a Lagrange multiplier that sets the "electromagnetic" field to zero,

$$S[\phi, \tilde{\phi}, A] = \frac{\beta^2}{2} \int d^2x \ \mathcal{D}_\mu \phi \mathcal{D}^\mu \phi + \frac{1}{2\pi} \int d^2x \ \tilde{\phi} \ \epsilon^{\mu\nu} \partial_\mu A_\nu \tag{3.14}$$

$\tilde{\phi}$ is called the *dual* field, and it is nothing but a Lagrange multiplier that sets $\epsilon^{\mu\nu}\partial_\mu A_\nu = F$ to zero. Therefore, naively,

$$\int \mathcal{D}\phi \mathcal{D}A \ \mathcal{D}\tilde{\phi} \ \ e^{iS[\phi,\tilde{\phi},A]} \overset{?}{=} Z(\beta) \quad \text{(as in (3.1))} \tag{3.15}$$

[2]The derivation is not really intuitive in the sense that I do not feel like rediscovering it as I am learning. Perhaps there is a simpler, more "intuitive" derivation, that I am not aware of.

This is actually very subtle. It turns out that the equality holds only if the dual boson $\tilde{\phi}$ is also compact, with a period 2π. To understand why, let us take a step back and count the degrees of freedom in the action (3.1) and (3.14). The original boson had one propagating degree of freedom, which is the boson that obey the wave equation. On top of that, it has winding degrees of freedom, labeled by $Q_{\text{wind}} \in \mathbb{Z}$.

On the other hand, it seems like introducing the gauge field seems to increase the number of degrees of freedom. However, we can choose a gauge such that the boson is zero everywhere,

$$\phi^{\text{new}} = \phi^{\text{old}} + \lambda = 0 \tag{3.16}$$

This is called the unitary gauge[3]. We are then left with the gauge field only, and in 2 spacetime dimensions, it has no propagating degree of freedom. The only candidate left in (3.14) to be a propagating degree of freedom is the dual field. But at this stage, it does not seem to have any dynamics. But integrating out the dual field will set $F = 0$, which means that we can choose $A = 0$ at least locally, and therefore keep the one local propagating degree of freedom ϕ intact. This suggests that if we gauge fix ϕ to zero, then the local propagating degree of freedom is hiding in the dual field, and we will soon see that this is indeed the case.

What about the windings? Let us use the unitary gauge again. Recall that the covariant derivative is invariant,

$$\mathcal{D}_\mu \phi^{\text{new}} = \mathcal{D}_\mu \phi^{\text{old}} \implies A_\mu^{\text{new}} = \partial_\mu \phi^{\text{old}} + A_\mu^{\text{old}}$$

Let $A_\mu^{\text{old}} = 0$. Then,

$$A_\mu^{\text{new}} = \partial_\mu \phi^{\text{old}} \tag{3.17}$$

Therefore, the winding of the original boson is encoded as holonomies in the unitary gauge,

$$Q_{\text{wind}} = \frac{1}{2\pi} \oint dx^\mu \partial_\mu \phi = \frac{1}{2\pi} \oint dx^\mu A_\mu \in \mathbb{Z} \tag{3.18}$$

Recall that holonomy is the mathematics terminology for what physicists would refer to as the Aharanov-Bohm phase. If the manifold in which the quantum particle is moving have loops that cannot be shrunken down to zero, then the particle can pick up a non-trivial observable phase $e^{i \oint_\gamma A}$ as it traverse that loop. In our case, due to the windings of the boson is quantized, these holonomies must also be quantized as shown in (3.18).

[3] Recall that $e^{i\phi}$ is the sensible field operator. Then setting ϕ to zero is like the unitary gauge that we have seen for a complex field, and hence the name.

How do we ensure that in (3.14), the holonomies are quantized? Let us work in the unitary gauge, where the action looks like,

$$S[A,\tilde{\phi}] = \frac{\beta^2}{2}\int d^2x\; A_\mu A^\mu + \frac{1}{2\pi}\int d^2x\;\tilde{\phi}\;\epsilon^{\mu\nu}\partial_\mu A_\nu \tag{3.19}$$

Here, integrating out $\tilde{\phi}$ imposes $F=0$. Of course, it does not mean that $A=0$ globally. It indeed allows for non-trivial holonomies where $\oint A \neq 1$ around a non-contractible cycle. But this is not enough. We need to ensure that these holonomies are "trivial", which means that the quantization condition in (3.18) is obeyed[4]. To this end, we will choose the dual field also to be compact, that is,

$$\tilde{\phi} + 2\pi\tilde{\beta} \sim \tilde{\phi} \tag{3.20}$$

We will also put the theory in finite temperature, which makes the temporal coordinate take values in $[0,\frac{1}{T}]$, where T is the temperature. Having $\tilde{\phi}$ compact, it can wind around cycles of the manifold. In our example of finite temperature field, the manifold is a torus $S^1(R)\times S^1(\frac{1}{T})$, which means that there are two non contractible cycles in the manifold, one around spatial part, $S^1(R)$ and the other around the temporal part $S^1(\frac{1}{T})$. Consider the path integral,

$$\int \mathcal{D}\tilde{\phi}\; e^{\frac{i}{2\pi}\int \tilde{\phi}F} \tag{3.21}$$

Integrating by parts, we have

$$\int \tilde{\phi}F = -\int d\tilde{\phi}\wedge A \tag{3.22}$$

Now the periodicity of the dual field imply that it can wind,

$$\oint_\gamma d\tilde{\phi} = 2\pi m_\gamma \tilde{\beta}, \quad \text{where} \quad m_\gamma \in \mathbb{Z} \tag{3.23}$$

Hence,

$$\int d\tilde{\phi}\wedge A \supset 2\pi m_\tau \tilde{\beta}\oint_\sigma A + 2\pi m_\sigma \tilde{\beta}\oint_\tau A \tag{3.24}$$

Then, the path integral over the dual field should also involve sums over these windings, to take into account all the winding sectors. We will therefore have,

$$\sum_{m_\sigma,m_\tau}\exp\left(im_\sigma\tilde{\beta}\oint_\tau A + im_\tau\tilde{\beta}\oint_\sigma A\right) = (2\pi)^2\sum_{k_\tau,k_\sigma}\delta\left(\oint_\tau A - \frac{2\pi k_\sigma}{\tilde{\beta}}\right)\delta\left(\oint_\sigma A - \frac{2\pi k_\tau}{\tilde{\beta}}\right) \tag{3.25}$$

Therefore, to make the holonomies be trivial (i.e, obey (3.18)), all we have to do is to tune the radius of the dual boson $\tilde{\beta}$ to equal 1.

[4]Indeed, this means that the Aharanov-Bohm phase around this loop is simply 1.

In summary, for (3.14) to be truly same as the theory in (3.1), we need the dual boson to be also compact with a radius equal to 1.

We are now ready to derive the duality. We saw that integrating out $\tilde{\phi}$ from (3.14) gives the original boson theory, (3.1). Now we will reverse the order of integration, and integrate out ϕ and A.

$$\begin{aligned} Z(\beta) &= \int \mathcal{D}\phi \ \exp\Big(i \int d^2x \ \frac{1}{2}\beta^2 \partial_\mu \phi \partial^\mu \phi\Big) \\ &= \int \mathcal{D}\phi \mathcal{D}A \ \mathcal{D}\tilde{\phi} \ \exp\Big[i\frac{\beta^2}{2} \int d^2x \ \mathcal{D}_\mu \phi \mathcal{D}^\mu \phi + \frac{i}{2\pi} \int d^2x \ \tilde{\phi} \ \epsilon^{\mu\nu} \partial_\mu A_\nu\Big] \\ &= \int \mathcal{D}\tilde{\phi} \mathcal{D}A \ \exp\Big[i\frac{\beta^2}{2} \int d^2x \ A_\mu A^\mu + \frac{i}{2\pi} \int d^2x \ \tilde{\phi} \ \epsilon^{\mu\nu} \partial_\mu A_\nu\Big] \\ &= \int \mathcal{D}\tilde{\phi} \ \exp\Big(i \int d^2x \ \frac{1}{2}\Big(\frac{1}{4\pi^2\beta^2}\Big) \partial_\mu \tilde{\phi} \partial^\mu \tilde{\phi}\Big) \\ &= Z(\tfrac{1}{2\pi\beta}) \end{aligned} \tag{3.26}$$

That is, the compact boson with radius β is completely equivalent to a compact boson with radius $\frac{1}{2\pi\beta}$. This is called the T-duality.

Aside : What is going on?

In the context of string compactification, T-duality would mean that the string cannot distinguish between a very small compact dimension and a very big compact dimension. This sounds a bit crazy. So it is worthwhile spending some extra time to understand what is going on, albeit the discussion being a tangent to the main focus of the notes. The equation of motion of the boson can be solved as,

$$\phi = \phi_{\text{zero-mode}} + \phi_{\text{osc}} \tag{3.27}$$

We will discard the oscillatory modes for this discussion. Recall that we have the boundary condition,

$$\phi(x + 2\pi R, \tau) = \phi(x, \tau) + 2\pi m \tag{3.28}$$

The zero mode therefore must take the form,

$$\phi = \phi_0 + a\tau + \frac{m}{R}x \tag{3.29}$$

m is the winding number. The total momentum of the field[5] is,

$$p = \int_{S^1(R)} dx\ \beta^2 \partial_\tau \phi = 2\pi R \beta^2 a \tag{3.30}$$

Since the coordinate ϕ is periodic, the total momentum must be quantized in integer units. The easiest way to see this is to look at the wavefunctional for the field, and it will involve $e^{ip\phi}$. For the wave functional to be single valued, we must have,

$$p = n \in \mathbb{Z} \tag{3.31}$$

This tells us that,

$$a = \frac{n}{2\pi R\beta^2} \implies \phi = \phi_0 + \frac{n}{2\pi R\beta^2}\tau + \frac{m}{R}x \tag{3.32}$$

The energy of the field configuration[6] is,

$$E = \int_{S^1(R)} dx\ \frac{1}{2}\Big(\beta^2\dot{\phi}^2 + \beta^2\phi'^2\Big) = \frac{\pi}{R}\left(\frac{n^2}{(2\pi\beta)^2} + \beta^2 m^2\right) \tag{3.33}$$

We now see T-duality in a different way. The spectrum stays invariant if we swap $\beta \leftrightarrow \frac{1}{2\pi\beta}$ and $m \leftrightarrow n$ at the same time. That is, T-duality exchanges the momentum modes with the winding modes.

Now, if the radius of the compact direction β is really small, the momentum modes become really high energy and becomes irrelevent for the IR physics. But, the winding modes become important. At large radius β, the opposite happens. This is how strings get confused about the size of the compact dimension. The string has two modes, which are equivalent under T-duality, and they take turns to be relevant for the IR physics as we change the radius.

We will take a moment to understand how the swapping of winding and momentum modes happen from the point of view of conserved currents. The momentum and winding modes of the boson are the shift and wind currents that we defined in (3.3) and (3.4). Let us start from the action after gauge fixing the boson to zero in (3.19). Recall that here the choice of gauge is,

$$A_\mu = \partial_\mu \phi \tag{3.34}$$

[5]Recall the canonical conjugate momentum is $\Pi = \beta^2\dot{\phi}$. The total momentum is the zero one component of the energy momentum tensor that we derive from worldsheet translational symmetry.

[6]Find the Hamiltonian or find the energy momentum tensor, and take the zero zero component, and then integrate over space.

where ϕ is the boson before gauge fixing as in (3.14). If we now vary (3.19) with respect to A, we get the equation of motion,

$$A_\mu = \frac{1}{2\pi\beta^2}\epsilon_{\mu\nu}\partial^\nu\tilde{\phi} \tag{3.35}$$

This gives us,

$$\beta^2\partial_\mu\phi = \frac{1}{2\pi}\epsilon_{\mu\nu}\partial^\nu\tilde{\phi} \tag{3.36}$$

which is,

$$j^\mu_{\text{shift}} = \tilde{j}^\mu_{\text{wind}} \tag{3.37}$$

This is the swapping of shift and wind currents under T-duality.

An Alternate, Quick Derivation

There is a quicker way of bypassing most of the details in the previous derivation. We start with the path integral (3.10). Since the boson only appear as a derivative in the path integral, we can equivalently integrate over all possible $d\phi$'s.

$$\int \mathcal{D}(\partial\phi)\ \exp\left(i\int_M d^2x\frac{\beta^2}{2}\partial_\mu\phi\partial^\mu\phi\right) \tag{3.38}$$

This is like integrating over F instead of A for electromagnetism. However, $F = d\phi$ cannot be any random function, it must be a closed function, $dF = 0$. We have to restrict the path integral over $d\phi$ to only closed functions. This is easily done by including a Lagrange multiplier $\tilde{\phi}$,

$$\int \mathcal{D}F_\mu\mathcal{D}\tilde{\phi}\ \exp\left(i\int_M d^2x\frac{\beta^2}{2}F^2 + \frac{i}{2\pi}\int_M d^2x\ \partial_\mu\tilde{\phi}\epsilon^{\mu\nu}F_\nu\right) \tag{3.39}$$

Integrating by parts the last term, and integrating out $\tilde{\phi}$ will do what we want, set $dF = \epsilon^{\mu\nu}\partial_\nu F_\mu = 0$. But this is too restrictive. We want to keep the winding modes of ϕ. By now, we know what to do. We just need to make $\tilde{\phi}$ also periodic. Then, summing over the winding sectors of $\tilde{\phi}$ while doing the path integral will ensure that F obey,

$$\frac{1}{2\pi}\int dx^\mu F_\mu \in \mathbb{Z} \tag{3.40}$$

As before, we have to tune the radius of $\tilde{\phi}$ to 1 to achieve this. The rest is the same, integrate out F, and discover the dual boson action!

3.3 Quantization

We will now quantize the compact boson. To this end, we will introduce chiral bosons. Notice that the equation of motion $\partial^2\phi = 0$ can be solved by

$$\phi = \phi_-(x^-) + \phi_+(x^+) \tag{3.41}$$

where $x^\pm = t \pm x$. This is almost true, apart from a zero mode that we will ignore for now. We can also write the dual boson in terms of the chiral modes. We have the following relation between the boson and the dual,

$$\beta^2 \partial^\mu \phi = \frac{1}{2\pi} \epsilon^{\mu\nu} \partial_\nu \tilde{\phi} \tag{3.42}$$

We can now try $\mu = \pm$, and that gives

$$\beta^2 \partial_- \phi_- = \frac{1}{2\pi} \partial_- \tilde{\phi}, \quad \text{and} \quad \beta^2 \partial_+ \phi_+ = -\frac{1}{2\pi} \partial_+ \tilde{\phi} \tag{3.43}$$

This is solved by

$$\tilde{\phi} = 2\pi\beta^2 \left(\phi_- - \phi_+\right) \tag{3.44}$$

The chiral bosons therefore can be written in terms of the boson and the dual,

$$\phi_\pm = \frac{1}{2}\left(\phi \mp \frac{1}{2\pi\beta^2}\tilde{\phi}\right) \tag{3.45}$$

We will make use of this relation later on. Now we proceed with quantization. This is the standard procedure. We write the general solution for the field ϕ, and turn the coefficients of the modes into operators. We have

$$\phi(x) = \frac{1}{\beta} \int \frac{dp}{2\pi} \frac{1}{\sqrt{2|p|}} \left(a_p e^{ipx} + a_p^\dagger e^{-ipx}\right) e^{-\frac{|p|}{2\Lambda}} \tag{3.46}$$

The canonical momentum, $\Pi = \beta^2 \dot{\phi}$, is given by

$$\Pi(x) = -i\beta \int \frac{dp}{2\pi} \sqrt{\frac{|p|}{2}} \left(a_p e^{ipx} - a_p^\dagger e^{-ipx}\right) e^{-\frac{|p|}{2\Lambda}} \tag{3.47}$$

The Λ is a UV cut off. Imposing equal time commutation relations, we get

$$[\phi(x,t), \Pi(y,t)] = i\delta(x-y) \quad \Longrightarrow \quad \left[a_p, a_q^\dagger\right] = 2\pi\delta(p-q) \tag{3.48}$$

We now want to construct the mode expansion for the chiral bosons. For that, notice that the dual boson obeys

$$\partial_x \tilde{\phi} = -2\pi\beta^2 \partial_t \phi \quad \Longrightarrow \quad \tilde{\phi}(x,t) = -2\pi\beta^2 \int_{-\infty}^{x} dx' \frac{1}{\beta^2} \Pi\left(x', t\right) \tag{3.49}$$

Then, we have

$$\phi_\pm(x,t) = \frac{1}{2}\left(\phi(x,t) \pm \frac{1}{\beta^2}\int_{-\infty}^{x} dx'\Pi\left(x',t\right)\right) \tag{3.50}$$

This tells us that the chiral bosons are highly non-local. To find the chiral boson at some point x, we have to have the knowledge of the fundamental fields at all points to the left side of x. We can now plug in the mode expansions. We will use

$$\int_{-\infty}^{x} dx' e^{-ipx'} = \frac{e^{-ipx}}{-ip} \quad \text{where we ignored the contribution from } -\infty \tag{3.51}$$

Using this, we find

$$\phi_\pm(x,t) = \frac{1}{2\beta}\int \frac{dp}{2\pi}\frac{1}{\sqrt{2|p|}}\left(1 \pm \frac{|p|}{p}\right)\left(a_p e^{ipx} + a_p^\dagger e^{-ipx}\right)e^{-\frac{|p|}{2\Lambda}} \tag{3.52}$$

Therefore,

$$\begin{aligned}
\phi_+(x,t) &= \frac{1}{2\beta}\int_0^\infty \frac{dp}{2\pi}\sqrt{\frac{2}{p}}\left(a_p e^{ipx} + a_p^\dagger e^{-ipx}\right)e^{-\frac{p}{2\Lambda}} \\
\phi_-(x,t) &= \frac{1}{2\beta}\int_{-\infty}^0 \frac{dp}{2\pi}\sqrt{\frac{2}{|p|}}\left(a_p e^{ipx} + a_p^\dagger e^{-ipx}\right)e^{-\frac{|p|}{2\Lambda}}
\end{aligned} \tag{3.53}$$

We see that the chiral bosons are left movers or right movers indeed. We also work out the commutation relations,

$$\begin{aligned}
[\phi_+(x,t), \phi_\pm(y,t)] &= \frac{1}{4\beta^2}\left(\pm\int_{-\infty}^{y} dx'\left[\phi(x,t), \Pi\left(x',t\right)\right] + \int_{-\infty}^{x} dx'\left[\Pi\left(x',t\right), \phi(y,t)\right]\right) \\
&= \frac{i}{4\beta^2}\left(\pm\int_{-\infty}^{y} dx'\delta\left(x-x'\right) - \int_{-\infty}^{x} dx'\delta\left(x'-y\right)\right) \\
&= \frac{i}{4\beta^2}(\pm\Theta(y-x) - \Theta(x-y))
\end{aligned} \tag{3.54}$$

In the second to third step, for example, the first integral gives 1 as the answer only if $y > x$. In short, we have

$$\begin{aligned}
[\phi_\pm(x,t), \phi_\pm(y,t)] &= \mp\frac{i}{4\beta^2}\,\mathrm{sign}(x-y) \\
[\phi_+(x,t), \phi_-(y,t)] &= \begin{cases} -\frac{i}{4\beta^2}, & \text{when } x \neq y \\ -\frac{i}{2\beta^2}, & \text{when } x = y \end{cases}
\end{aligned} \tag{3.55}$$

We have defined the Heavyside function to be unity when both arguments coincide. Only this choice would make a later calculation sensible.

3.4 Correlation Functions

One important thing to notice is that the chiral bosons do not commute no matter how far apart they are. This again is another manifestation of the non-locality of the chiral bosons. Also, the left and right modes do not commute. This turns out to be (I don't know how!) a consequence of us ignoring the zero mode. We will not treat the zero mode carefully here. Let us compute the "connected" Green's function for the chiral bosons. It is defined as

$$G_{\pm}(x,y) = \langle \phi_{\pm}(x)\phi_{\pm}(y)\rangle - \left\langle \phi_{\pm}(0)^2 \right\rangle \tag{3.56}$$

The subtraction is to have the Green's function be non-divergent for short distances. We have

$$\begin{aligned} G_{+}(x,y) &= \frac{1}{4\beta^2}\int_0^\infty \frac{dp}{2\pi}\sqrt{\frac{2}{p}}\int_0^\infty \frac{dq}{2\pi}\sqrt{\frac{2}{q}}\left\langle a_p a_q^\dagger \right\rangle \left(e^{ipx-iqy}-1\right) e^{-\frac{p+q}{2\Lambda}} \\ &= \frac{1}{4\beta^2}\int_0^\infty \frac{dp}{2\pi}\frac{2}{p}\left(e^{ip(x-y)}-1\right)e^{-\frac{p}{\Lambda}} \\ &= \frac{1}{4\pi\beta^2}\log\left(\frac{\epsilon}{\epsilon+i(x-y)}\right), \quad \text{where } \epsilon = \frac{1}{\Lambda} \end{aligned} \tag{3.57}$$

Similarly,

$$G_{-}(x,y) = \frac{1}{4\pi\beta^2}\log\left(\frac{\epsilon}{\epsilon-i(x-y)}\right) \tag{3.58}$$

One important thing to remember is that ϕ (or the dual) is not a well-defined object, as it is not single valued. Therefore, the chiral bosons are not really the operators of interest. Instead, one should look at $\partial\phi$, or the so-called vertex operators,

$$: e^{i\phi} : \tag{3.59}$$

The $: \cdot :$ symbol means normal ordering, where the creation operators are all brought to the left side. In the remaining, we will simply omit the normal ordering symbol when there is no confusion. Now, we will compute correlation functions of the type

$$\left\langle : e^{i\phi_-} :: e^{-i\phi_-} : \right\rangle \quad \text{and} \quad \left\langle : e^{i\phi_+} :: e^{-i\phi_+} : \right\rangle \tag{3.60}$$

One important question is whether $: e^{-i\phi_-} :$ obeys the periodicity conditions of ϕ. We will take this up later. Now, we will compute them. For that, we will need the BCH formula. Let A, B be operators with $[A,B] = $ constant. Then,

$$e^A e^B = e^{A+B} e^{\frac{1}{2}[A,B]} = e^B e^A e^{[A,B]} \tag{3.61}$$

For $A = \alpha a + \alpha' a^\dagger$ and $B = \beta a + \beta' a^\dagger$, we have

$$\begin{aligned}
:e^A::e^B: &= e^{\alpha' a^\dagger} e^{\alpha a} e^{\beta' a^\dagger} e^{\beta a} \\
&= e^{\alpha' a^\dagger} e^{\beta' a^\dagger} e^{\alpha a} e^{\beta a} e^{\alpha \beta'} \\
&=: e^{A+B} : e^{\langle AB \rangle}
\end{aligned} \tag{3.62}$$

The expectation value of $: e^{A+B} :$ is 1. Therefore,

$$\Big\langle : e^A :: e^B : \Big\rangle = e^{\langle AB \rangle} \tag{3.63}$$

This gives us

$$\begin{aligned}
\Big\langle : e^{i\phi_-(x)} :: e^{-i\phi_-(y)} : \Big\rangle = e^{G_-(x-y)} = \left(\frac{\epsilon}{\epsilon - i(x-y)} \right)^{\frac{1}{4\pi\beta^2}} \\
\Big\langle : e^{i\phi_+(x)} :: e^{-i\phi_+(y)} : \Big\rangle = e^{G_+(x-y)} = \left(\frac{\epsilon}{\epsilon + i(x-y)} \right)^{\frac{1}{4\pi\beta^2}}
\end{aligned} \tag{3.64}$$

In CFT language, we would say that the dimension (i.e., the scaling dimension) of the vertex operators is $\frac{1}{8\pi\beta^2}$.

4 Bosonization

In this section, we will see how the free fermion theory and compact boson theory are mapped to each other in two ways. First from the point of view of the correlation functions, and then the point of view of the path integral.

4.1 Bosonization from Correlators

The claim is the following: for a magic value of the radius of the compact boson $\beta^2 = \frac{1}{4\pi}$, the correlation functions of the free fermion and the compact boson vertex operators are identical, upto some constants. Let us make a table, The identification is,

Fermion	**Boson**
$\langle \psi_-^\dagger(y)\psi_-(x)\rangle = \frac{i}{2\pi}\frac{1}{(y-x)+i\epsilon}$	$\langle : e^{i\phi_-(x)} :: e^{-i\phi_-(y)} :\rangle = \frac{\epsilon}{\epsilon - i(x-y)}$
$\langle \psi_+^\dagger(y)\psi_+(x)\rangle = \frac{i}{2\pi}\frac{1}{(y-x)+i\epsilon}$	$\langle : e^{i\phi_+(x)} :: e^{-i\phi_+(y)} :\rangle = \frac{\epsilon}{\epsilon - i(x-y)}$

$$\psi_-(x) \longleftrightarrow \sqrt{\frac{1}{2\pi\epsilon}} e^{i\phi_-(x)} \tag{4.1}$$

Similarly,

$$\psi_+(x) \longleftrightarrow \sqrt{\frac{1}{2\pi\epsilon}} e^{-i\phi_+(x)} \tag{4.2}$$

We can map composite operators as well. For example, the mass term,

$$:\bar{\psi}\psi :=: \psi_-^\dagger(x)\psi_+(x) : + : \psi_+^\dagger(x)\psi_-(x) : \quad \longleftrightarrow \frac{1}{2\pi\epsilon}\left(: e^{-i\phi_-(x)}e^{-i\phi_+(x)} : + : e^{i\phi_+(x)}e^{i\phi_-(x)} :\right) \tag{4.3}$$

Using the BCH formula,

$$e^{-i\phi_-(x)}e^{-i\phi_+(x)} = e^{-i(\phi_-(x)+\phi_+(x))}e^{-\frac{1}{2}[\phi_-(x),\phi_+(x)]} = e^{-i\phi(x)}e^{-\frac{1}{2}\frac{-i}{4\left(\frac{1}{2\pi}\right)}} = e^{-i\phi(x)}e^{i\pi} = -e^{-i\phi(x)} \tag{4.4}$$

Here, we used

$$[\phi_-(x), \phi_+(x)] = \frac{-i}{2\beta^2} \tag{4.5}$$

This came from defining the Heaviside function to have the value one at the coinciding point. Therefore, we see that the mass term in the fermion theory is mapped to,

$$:\bar{\psi}\psi : \longleftrightarrow -\frac{1}{2\pi\epsilon}\left(e^{-i\phi(x)} + e^{i\phi(x)}\right) = -\frac{1}{\pi\epsilon}\cos(\phi) \tag{4.6}$$

It can be shown that the chiral mass term,

$$i\bar{\psi}\gamma^5\psi \longleftrightarrow -\frac{1}{\pi\epsilon}\sin(\phi) \tag{4.7}$$

Matching Currents

We have matched the operators between the two theories. Now we will match the currents. We will see that the vector current of one theory is mapped to the topological current of the other theory. The vector current in the fermionic theory was,

$$j_V^\mu = \bar{\psi}\gamma^\mu\psi \Longrightarrow \begin{cases} j_V^0 = \psi_+^\dagger\psi_+ + \psi_-^\dagger\psi_- \\ j_V^1 = -\left(\psi_+^\dagger\psi_+ - \psi_-^\dagger\psi_-\right) \end{cases} \tag{4.8}$$

If we define these composite operators naively, we will get infinities. We will now define them carefully, and comment on the naive definition in that light. We define them by point splitting.

$$\begin{aligned} \psi_-^\dagger\psi_- &:= \lim_{x\to y}\psi_-^\dagger(x)\psi_-(y) \\ &\leftrightarrow \lim_{x\to y}\frac{1}{2\pi\epsilon} : e^{-i\phi_-(x)} :: e^{i\phi_-(y)} : \\ &= \lim_{x\to y}\frac{1}{2\pi\epsilon} : e^{-i(\phi_-(x)-\phi_-(y))} : e^{G_-(x,y)} \\ &= \lim_{x\to y}\frac{1}{2\pi\epsilon} : e^{-i(\phi_-(x)-\phi_-(x)-(y-x)\partial_x\phi_-(x)-\dots)} : e^{\log\left(\frac{\epsilon}{\epsilon-i(x-y)}\right)} \\ &= \lim_{x\to y}\frac{1}{2\pi\epsilon} : (1-i(x-y)\partial_x\phi_-(x)-\dots) : \frac{\epsilon}{\epsilon-i(x-y)} \\ &= \frac{1}{2\pi} : \partial_x\phi_-(x) : + \lim_{x\to y}\frac{i}{2\pi(x-y)} \end{aligned} \tag{4.9}$$

We see that the correlation function blows up. Had we not defined the composite operator via point splitting, we will not get the finite term. We will define the fermionic composite operator to be normal ordered, which means we will ignore this infinite part. This leaves us with,

$$: \psi_\pm^\dagger\psi_\pm : \longleftrightarrow \frac{1}{2\pi} : \partial_x\phi_\pm : \tag{4.10}$$

Then the fermionic vector current in the bosonized version reads,

$$\begin{aligned} j_V^0 &= \frac{1}{2\pi}\partial_x(\phi_+ + \phi_-) = \frac{1}{2\pi}\partial_x\phi \\ j_V^1 &= -\frac{1}{2\pi}\partial_x(\phi_+ - \phi_-) = -\frac{1}{2\pi\beta^2}\Pi = -\frac{1}{2\pi}\partial_t\phi \end{aligned} \tag{4.11}$$

In short,

$$j_V^\mu \longleftrightarrow \frac{1}{2\pi}\epsilon^{\mu\nu}\partial_\nu\phi \tag{4.12}$$

The vector current of the fermionic theory is mapped to the topological current of the bosonic theory. We can show that the topological current (i.e., the axial current) of the fermionic theory is mapped to the vector current (i.e., the shift current) of the bosonic current.

A Subtlety

We already mentioned that only operators that respect the periodicity of the boson are sensible. For example, $e^{in\phi}$ and $e^{im\phi}$ are both sensible operators when m, $n \in \mathbb{Z}$. We should then ask the question if $e^{i\phi_\pm}$, the operators that we used to construct the bosonization map are sensible. Let us write down a general, sensible exponential type operator in the bosonic theory. That would look like,

$$e^{in\phi+im\tilde{\phi}} \tag{4.13}$$

Writing the boson $\phi = \phi_+ + \phi_-$, and $\tilde{\phi} = 2\pi\beta^2\,(\phi_- - \phi_+)$,

$$e^{i\left(n-2\pi\beta^2 m\right)\phi_+ + i\left(n+2\pi\beta^2 m\right)\phi_-} = e^{i\left(n-\frac{m}{2}\right)\phi_+ + i\left(n+\frac{m}{2}\right)\phi_-} \tag{4.14}$$

In the second line, we put in the specific value of β. To get a purely chiral operator, one must choose $n = 1, m = \pm 2$. This leaves us with one of $e^{i2\phi_\pm}$. We see that it is not possible to construct $e^{i\phi_\pm}$ in such a way that the periodicity of the boson and the dual are kept intact. Therefore, $e^{i\phi_\pm}$ are not in the spectrum of the theory. If the bosonization dictionary is believed to be one-to-one, one must also ensure that $\psi_\pm$ are also not in the spectrum of the fermionic theory. This turns out to be the case. I do not understand this completely. The idea is that the fermionic theory should have a Z_2 gauge symmetry. (I do not really know what it means!) Somehow it eliminates (I guess gauge symmetry eliminates some degrees of freedom, but don't really see how in this present case!) ψ, and leaves out only (perhaps even) composite operators like $\psi\psi$.

This begs another question. Suppose I chose $n = m = 1$. Then, we will get fractional powers of $e^{i\phi_\pm}$. What does this mean? What is the corresponding operator in the fermionic side? I don't know. Perhaps I should email David Tong!

References